

WEEK07: GRAPH TRAVERSALS

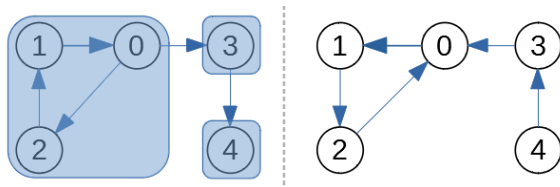
8.1 Strongly Connected Components

One of the multiple practical applications of a DFS traversal of a directed graph is finding strongly connected components (strongly connected graphs are defined in (Goodrich2011, p.626)), the relevant algorithm is known as Kosaraju's algorithm. <https://bit.ly/3II20ec>, <https://bit.ly/3mNU2la>.

Definition: A subset of vertices in a directed graph $S \subseteq G.V$ makes a strongly connected component, iff for any two distinct vertices u, v there is a path $u \rightsquigarrow v$ (one or more and also another path $v \rightsquigarrow u$ that goes back from v to u).

If you can travel only in one direction (say, from u to v), but cannot return, then u, v should be in different strongly connected components. (Same thing, if u and v are mutually unreachable.) Moreover, every vertex is strongly connected to itself – so even in the worst case a graph with n vertices would have at most n strongly connected components (containing one vertex each).

Figure shows an example of a graph with $n = 5$ vertices having 3 strongly connected components. Next to that graph is the *transposed graph* G^T where all the edges are reversed.



8.2 Kosaraju's algorithm

There is a way to find strongly connected components in an arbitrary graph by running DFS twice (i.e. it works in linear time $O(n + m)$).

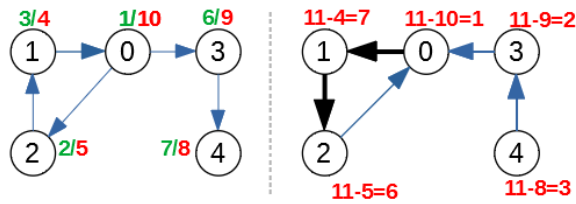
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STRONGLY_CONNECTED( $G$ )
  (compute all finishing times u.f)
1  call DFS( $G$ )
  ( $G^T$  is transposed  $G$ , all edges reversed)
2  compute  $G^T$ 
  (visit vertices in decreasing u.f order)
3  call DFS( $G^T$ )
4  for each tree  $T$  in the forest DFS( $G^T$ )
5    Output  $T$  as a component

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To see how this works, we can run it on the example graph shown earlier. After the DFS on graph G is run, we get the finishing times for the vertices 0, 1, 2, 3, 4 (all shown in red on the left side of Figure below). After that we replace G

by G^T (to the right side of the same figure), and assign priorities in the decreasing sequence of $u.f$ (the finishing times when running $\text{DFS}(G)$).



To make this reverse order obvious, we assign new priorities to the vertices in G^T . The new priorities in G^T are the following:

- Vertex 0 has priority $11 - 10 = 1$.
- Vertex 1 has priority $11 - 4 = 7$.
- Vertex 2 has priority $11 - 5 = 6$.
- Vertex 3 has priority $11 - 9 = 2$.
- Vertex 4 has priority $11 - 8 = 3$.

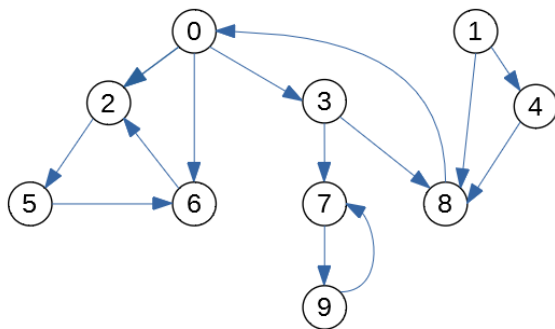
Now run $\text{DFS}(G^T)$. It turns out that the DFS algorithm starts in the vertex "0" once again (since it was finished last in $\text{DFS}(G)$). But unlike the DFS algorithm in G itself (it produced just one DFS tree), we get a DFS forest with 3 components (tree/discovery edges shown bold and black in the previous Figure).

- $\{0, 1, 2\}$ (DFS tree has root "0").
- $\{3\}$ (DFS tree has root "3").
- $\{4\}$ (DFS tree has root "4").

They represent the strongly connected components in G (they are also strongly connected in G^T).

8.3 Problem

We start with the graph shown in Figure below.



- Run the DFS traversal algorithm on the graph G . Mark each vertex with the pair of numbers d/f , where the first number d is the discovery time, and the second number f is the finishing time.
- Draw the transposed directed graph (same vertices, but each arrow points in the opposite direction). Run the DFS traversal algorithm on G^T . Make sure that the DFS outer loop visits the vertices in the reverse order by $u.f$ (the finishing time for the DFS algorithm in step (A)). In this case you do not produce the discovery/finishing times

once again, just draw the discovery edges used by the DFS on G^T – you can highlight them (show them in bold or use a different color).

- (C) List all the strongly connected components (they are the separate pieces in the forest obtained by running DFS on G^T).